

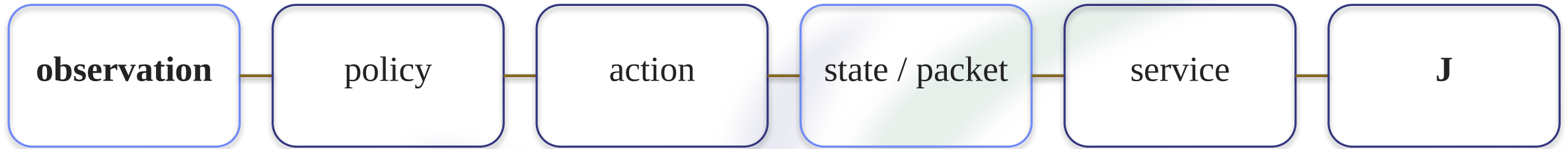
每一次 SEND 、 WAIT 、 SLEEP 都是能源決策



SIMULATED TEACHING DATA | 先追機制，再讀證據

× 智慧節能與物聯網應用

節能是一條可檢驗的因果鏈



每一段都要能回到 result / replay ；沒有 state 、 packet 、 service 或 energy 差異，
就不能稱為節能證據。

觀察先於選擇 | 服務先於 J

VERIFIED

LoRaEnergySim 為什麼直接服務智慧節能與 IoT



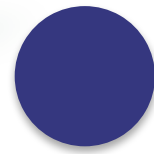
STATE

sleep / processing / TX /
RX



PACKET

collision / retry / delivered



ENERGY

state time $\rightarrow$ *endpoint J*

一個 action 會讓哪一層改變？先說機制，再看結果。

VERIFIED

LEO 只提供會改變的服務窗口

LEO = changing service window ; 本頁只示範窗口機制



窗口長短改變機會；機會改變 policy ； policy 改變 service / endpoint J

VERIFIED

兩層 evidence 不能互相冒充

ENDPOINT LAYER

queue
packet
state
endpoint J

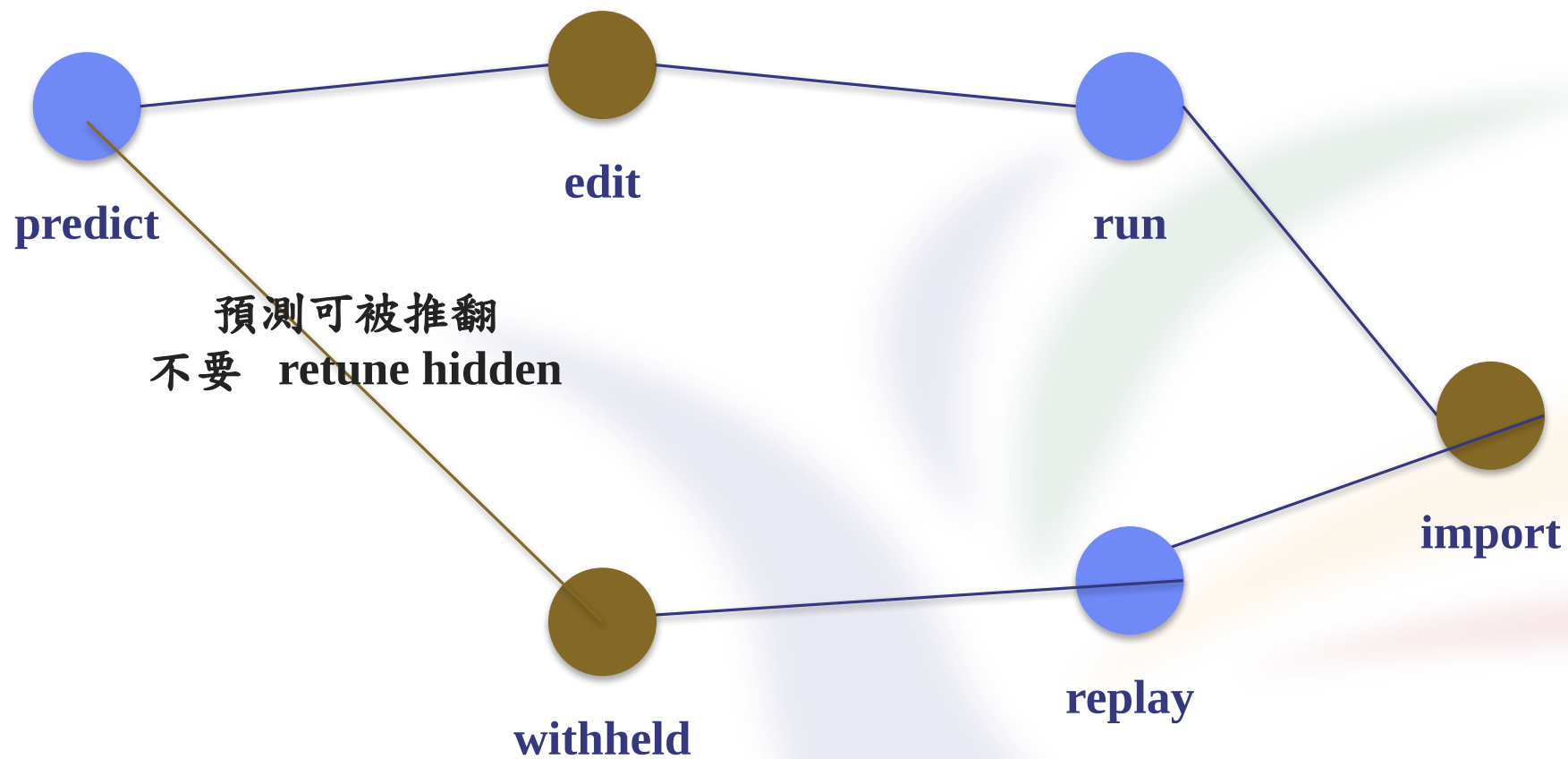
SYSTEM LAYER

既有 C-120 authority
canonical fields
不可互寫

共享 scenario / clock / workbook ，不共享語義與 energy boundary

VERIFIED

從預測走到可反駁的結果



VERIFIED

一個小改動，必須能追到一個機制

student_policy.py

MARKED BLOCK

只改這裡

policy hash → result → replay



scenario



schemas



runner

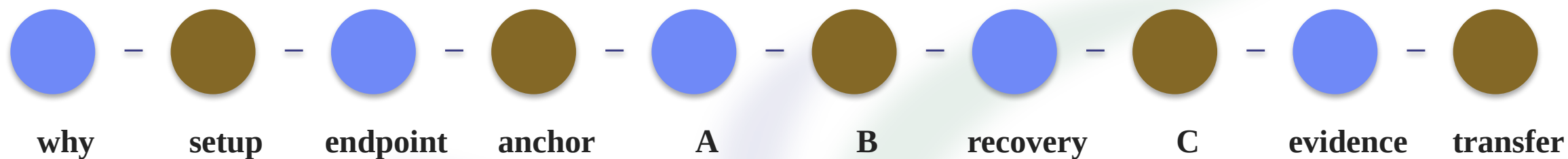


generated JSON

沒有 consequential diff → 停在 gate

VERIFIED

今天的教學路徑



每一個節點都要有 gate 、 receipt 與 recovery ；受阻時沿同一 scenario 回來。

VERIFIED

開場檢查點：先確認 claim 邊界

course-packaged simulated endpoint-energy lab

可說 course-owned / deterministic / simulated

不可說 NOT LIVE / NOT MEASURED

尚未驗證 NOT CANONICAL-PARITY-VERIFIED

處理方式 URL / screenshot / KPI 缺口 → PLACEHOLDER

邊界會跟著 result 、 replay 、 import 與 workbook provenance 一起走

VERIFIED

套件根目錄是一條 provenance 邊界

解壓後：cd lora-energy-lab
POSIX：find . -maxdepth 2 -type f

Windows：Get-ChildItem -Recurse -File

lora-energy-lab/

- └─ README.en.md / README.zh-TW.md
- └─ setup.sh / setup.cmd
- └─ course.sh / course.cmd
- └─ *student_policy.py*
- └─ lora_energy_lab/
- └─ scenarios/
- └─ schemas/

provenance gate

single root
reviewed bytes
commit 32cc723...
required files present
identity mismatch → STOP

VERIFIED LOCAL ARTIFACT

先核對 local artifact ，再等待公開 release

```
ls -l ~/lora-energy-lab-v1.zip  
sha256sum ~/lora-energy-lab-v1.zip
```

public URL: RELEASE_ASSET_PENDING

LOCAL ZIP

lora-energy-lab-v1.zip
121,139 bytes

SHA-256

047e8459988d8bee
59fbfdcc39189e00048f3697968901ce5b6e5
363dc9e8c8c

repo : github.com/cedarwud/lora-energy-lab

receipt : VERIFIED LOCAL ARTIFACT | 19/19 tests PASS

公開 GitHub Release URL : RELEASE_ASSET_PENDING

VERIFIED LOCAL ARTIFACT

Windows 原生先確認 shell 與 launcher

CMD

C:\...>

py --list
py -3.11 --version

PowerShell

PS C:\...>

py --list
py -3.11 --version

預期：PLACEHOLDER | 看到 Python 3.11.x 才進入 setup

IMPLEMENTED / NOT VERIFIED

沒有 Python 3.11 時的 Windows 安裝路徑

開啟：<https://www.python.org/downloads/windows/>
安裝後：`py --list`

確認：`py -3.11 --version`

1

official installer / uv

取得 Python 3.11.x

2

Python Launcher

保留 `py --list`

3

reopen shell

清掉舊 PATH 狀態

4

version gate

`py -3.11 --version`

IMPLEMENTED / NOT VERIFIED

uv 是 Python 3.11 的 recovery 工具

PowerShell / POSIX command ribbon

```
winget install --id=astral-sh.uv -e
```

```
uv python install 3.11
```

```
uv python find 3.11
```

receipt / purpose

預期 receipt :
PLACEHOLDER | uv
fresh setup 尚未驗證 ;
成功時回傳實際
path 。

IMPLEMENTED / NOT VERIFIED

Windows 用 package-local .venv

```
PS> Get-ChildItem .venv\Scripts\python.exe
```

SYSTEM PYTHON

other project
shared packages

```
PS> .\.venv\Scripts\python.exe --version
```

PACKAGE .venv

.venv\Scripts\python.exe
package-local lock

不要共享 WSL .venv | 正常上課不必 activate , launcher 優先找 package-local venv

IMPLEMENTED / NOT VERIFIED

Windows 執行 setup 與 verify

Windows PowerShell / package root

```
$env:PYTHON_BIN = "py -3.11"
```

```
.\setup.cmd
```

```
.\course.cmd verify
```

machine-readable receipt

status: READY

Python 3.11.x

scenario_id

policy_api_version

目前： **READY /
PLACEHOLDER**

IMPLEMENTED / NOT VERIFIED

WSL 先確認它真的是 Ubuntu shell

PowerShell (系統管理員)

```
wsl --install -d Ubuntu  
wsl -l -v  
→ 進入 Ubuntu
```

Ubuntu shell

```
pwd  
uname -a  
→ Linux home path 才用 setup.sh
```

IMPLEMENTED / NOT VERIFIED

WSL 安裝工具與 uv

```
sudo apt update
sudo apt install -y curl unzip
```

```
curl -Lsf https://astral.sh/uv/install.sh | sh
uv --version && uv python find 3.11
```

1

official installer / uv

取得 Python 3.11.x

2

Python Launcher

保留 py --list

3

reopen shell

清掉舊 PATH 狀態

4

version gate

py -3.11 --version

IMPLEMENTED / NOT VERIFIED

WSL 解壓與建立 .venv

```
cp /mnt/c/.../lora-energy-lab-v1.zip ~/
unzip lora-energy-lab-v1.zip
cd lora-energy-lab
```

```
PYTHON_BIN=python3.11 bash setup.sh
bash course.sh verify
```

Windows Downloads
/mnt/c/Users/
<YourName>/
Downloads/

WSL Linux home

~/lora-course/

package root

lora-energy-lab/

local venv

.venv/bin/python

WSL 專用 .venv/bin/python
不要把 Windows 既有 venv 帶進 WSL

IMPLEMENTED / NOT VERIFIED

POSIX runner 的最小入口

PowerShell / POSIX command ribbon

```
PYTHON_BIN=python3.11 bash setup.sh
```

```
bash course.sh verify
```

receipt / purpose

預期 receipt :
PLACEHOLDER |
POSIX clean run 尚未驗
證；成功時回
READY 。

IMPLEMENTED / NOT VERIFIED

READY receipt 是下一步的入場券

bash course.sh verify

Windows : .\course.cmd verify

status	READY / PLACEHOLDER
Python	3.11.9
scenario_id	ntpu-energy-decision-01
policy_api_version	lora-energy-policy-v1
engine_mode	coherent-course-simulated-adapter

PLACEHOLDER

scenario_sha256: c157c76ed283074cdd17c1b689b83e168022abd2ca6ef8d9c337e69ab31a6593
 scenario_anchor_sha256: cfe84682a71c0c0ea09c273d4cd033c32818da63cf1695ace4363b9d789c50f9
 lock_sha256: f1d1fe3339137d56e4e9ea9d141b764eaa183a8f40933bad597d734bbf310197
 baseline_policy_sha256: 9917d2cde2e8c0eae8339b7eea2e358bf8577b4eb6aaf1e928674f6b8964ef68

setup 失敗不等於 energy result

bash course.sh verify

失敗時：保留原始 stdout / stderr

verify

READY

進入 runner
保留同一 identity

RECOVERABLE ERROR

保留 stdout / stderr
修最小邊界或 fallback

fallback 保留相同學習問題，但不冒充本機 policy execution

VERIFIED

可編輯檔案只有一個

POSIX : sed -n '1,220p' student_policy.py
Windows : Get-Content .\student_policy.py

git diff -- student_policy.py

student_policy.py

MARKED BLOCK

只改這裡

policy hash → result → replay



scenario



schemas



runner



generated JSON

沒有 consequential diff → 停在 gate

VERIFIED

policy API 的輸入是現在能觀察到的事

bash course.sh verify

Windows : .\course.cmd verify

observation

當下允許讀取的狀態
quality / queue / window

choose_action(...)

WAIT / SLEEP
SEND_ONE / SEND_URGENT
FLUSH_BATCH

不能偷看 future quality 、 future energy 或 result summary ；契約外 action 要 fail closed 。

PLACEHOLDER

Python survival : 常數是受控旋鈕

```
grep -n PACE_GAP_STEPS student_policy.py
```

```
bash course.sh verify
```

student_policy.py | marked constants

```
PACE_GAP_STEPS = bounded  
REST_DURING_GAP = WAIT / SLEEP  
BATCH_SIZE = bounded  
URGENT_MARGIN_S = bounded
```

bounded values only

先預測
只改一個
保留小 diff

PLACEHOLDER

Python survival : 條件只處理當前 observation

POSIX : `grep -n 'def choose_action' student_policy.py`

`bash course.sh verify`

現在看到的 condition

if

WAIT / SLEEP

elif

SEND_ONE

else

SEND_URGENT

每條分支只回傳一個 legal action ; 不要讀 future outcome 。

PLACEHOLDER

Python survival : 縮排與 return

```
python -m py_compile student_policy.py  
py -3.11 -m py_compile student_policy.py
```

```
bash course.sh verify
```

choose_action(observation)

```
if ready:  
    return WAIT  
elif low_power:  
    return SLEEP  
else:  
    return SEND_ONE
```

runner 收到什麼？

```
    縮排  
→ 分支歸屬  
    return  
→ action token
```

PLACEHOLDER